

Arduino Based ESP8266 WIRELESS WEB SERVER

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The project is based on low-cost ESP8266 Wi-Fi module. Currently, ESP8266 is gaining popularity in the field of electronics because it is low-cost, reliable and easily available in the market. Most documents related to this module are in Mandarin (Chinese language) and information provided in the data sheet is not sufficient enough for using ESP8266 for an application. In order to fill the gap, people from various countries have formed an ESP8266 community forum, which gives necessary details about programming and other related issues concerning this module.

ESP8266 contains a built-in 32-bit low-power CPU, ROM and RAM. It is a complete and self-contained Wi-Fi network solution that can carry software applications as a stand-alone device or connected with a microcontroller (MCU). The module has built-in AT Command firmware to be used with any MCU via COM port. Salient features of ESP8266 are:

- 802.11 b/g/n protocol
- Direct (P2P), soft-AP
- Integrated TCP/IP protocol stack
- Integrated PLL, regulators and power-management units
- +19.5dBm output power in 802.11b mode
- Supports antenna diversity
- Integrated low-power 32-bit MCU
- SDIO 2.0, SPI, UART
- Wireless SoC
- Has GPIO, I2C, ADC, SPI, PWM
- Maximum frequency is 80MHz
- 64k bytes of instruction RAM
- 96k bytes of data RAM
- 64k bytes of boot RAM
- RISC architecture

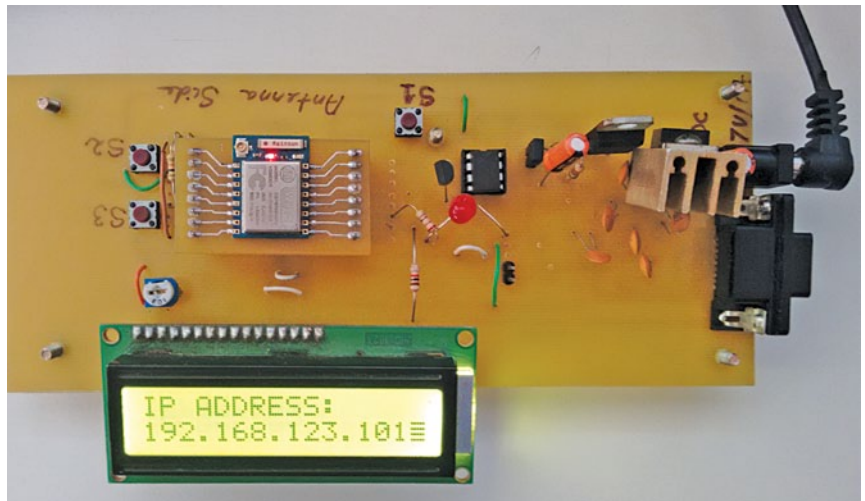


Fig. 1: Author's prototype



Fig. 2: ESP8266 module

Test Points

Test point	Details
TP1	+5V
TP2	+3.3V

(Voltage measured with respect to ground)

Circuit and working

Circuit diagrams of the main board and child board are shown in Figs 3 and 4, respectively. We have used ESP8266 as a stand-alone device. Supply voltage for the Wi-Fi module is 3.3 volts. Child board is to be

PARTS LIST

Semiconductors:

- MODULE1 - ESP8266 module (Sunrom part no. 4255)
- IC1 - 7805, 5V voltage regulator
- IC2 - LM317 adjustable voltage regulator
- IC3 - LM35 temperature sensor
- IC4 - LM358 operational amplifier
- IC5 - HCF4094 8-stage shift and store bus register
- IC6 - MAX3232 dual EIA-3232 driver/receiver
- LED1 - 5mm LED

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

- R1, R6 - 220-ohm
- R2 - 330-ohm
- R3-R5, R7, R8 - 10-kilo-ohm
- VR1 - 10-kilo-ohm potmeter

Capacitors:

- C1 - 100 μ F, 25V electrolytic
- C2, C4-C7 - 0.1 μ F ceramic disk
- C3 - 1 μ F, 25V electrolytic

Miscellaneous:

- S1-S3 - Tactile (push-button) switch
- LCD1 - 16 $\times$ 2 character module display
- CON1 - 2-pin connector
- CON2 - DB9 connector
- CON3, CON4 - 8-pin base connector for child board
- CON5, CON6 - 8-pin header connector for child board
- J1 - 2-pin jumper connector
- USB-to-serial converter
- 7V-12V DC regulated power supply